RESEARCH PROGRESS

5/7/24

Ronald Adomako

Advisors: Ahmed Elgawady, Xiaosong Du

LITERATURE REVIEW

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: Application of Machine Learning Models to Bridge Afflux Estimation **Author:** Piraei, R.; Niazkar, M.; Afzali, S. H.; Menapace, A. (2023) **Institution:** Department of Civil Engineering, Shiraz University, Iran	• Numerical Methods: conservation of momentum and energy. • Numerical methods need calibration. • Large dataset (O(m) large memory 💽) • Experimental methods are expensive (💰) and time consuming (⏰, O(n) large runtime)	• Regression Models (8) vs Artificial Neural Network (ANN), Genetic Programming (GP), & explicit equations. • 202 lab data + 66 field data (= 268 data instances) ➢ Support Vector Regression (SVR) 👎 ➢ Decision Tree Regressor (DTR) ➢ Random Forest Regressor (RFR) ❓ ➢ AdaBoost Regressor (ABR) 👎 ➢ Gradient Boost Regressor (GBR) ➢ eXtreme Gradient Boosting [XGBoost] for Regression (XGBR) ➢ Gaussian Process Regression (GPR) ➢ K-Nearest Neighbor (KNN) • MinMax Scalar for Normalization • Performance: RMSE, MAE with validation of test set variance of min and max in between train set variance of min and max.	• Machine learning out-perform traditional equations in bridge afflux (civil engineering) • XGBR and GPR good models. [discrete smoothing] • SVR bad model. [poor Non-linear, discrete simulation]	• Limitation is to afflux in arch bridges. • Does not account for debris (discrete noise) in environment.	• Great machine learning methodology for regression models. • Not many like research to make claim of whether machine learning regressor models is preferred for other civil engineering studies.

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: Machine Learning to Predict Properties of Fresh and Hardened Alkali-activated Concrete **Author:** Gomaa, E., Han, T., ElGawady, M., Huang, J., & Kumar, A. (2020) **Institution**: Department of Civil Engineering, Missouri University of Science and Technology, USA	• Complexity in comparing Alkali-Activated Concrete (AAC) to Portland Cement Concrete (PCC) using primarily random forest (RF) as a machine learning method. • **Motivation**: ACC has a lower impact on environment than PCC has on environment.	• 200 data points of AAC • Various compositions of AAC under different conditions to establish robust datasets. • Rigorous performance evaluation using Pearson's correlation coefficient, MAE, and RMSE • Classification-and-regression trees [CART] Random Forest (RF) to predict composition dependent slump flows and compressive strength of AAC	• RF accurately predicts slump flow of AAC. • Based on accuracy design mixture and conditions can be iterated then evaluated using same model.	• Comparison of RF model to physical model was not explicit.	• Study in material science evaluating machine learning with favorable result. • Only one machine learning method was used. • Accuracy was high with RF and optimization using XGBoost may be unnecessary, but can validate strength of RF.

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: Machine Learning Enabled Closed-Form Models to Predict Strength of Alkali-Activated Systems **Author:** Han, T., Gomaa, E., Gheni, A., Huang, J., ElGawady, M., & Kumar, A. (2022) **Institution**: Department of Civil Engineering, Missouri University of Science and Technology, USA	• Developing Alkali-Activated Mortars (AAM) using both conventional data and newly proposed chemo-structural descriptors. • Newly developed descriptors provide a novel approach to modeling the reactivity and properties of AAM by capturing both the chemical composition and the structural configuration of the raw materials.	• Random Forest (RF) on feature extraction for chemo-structural predictors for both network ratio and no. of constraints.	• Machine learning with chemo-structural predictors significantly outperform traditional models on chemical composition alone. • Insights on new mechanical properties offered. • Simplification of network ratio and no. of constraints improve interpretability and accuracy. • Significant predictors for compressive strength established.	• Experiment can be expanded to more than just AAM types.	• Innovative approach for new predictors. • Model depends heavily on extensive data-expensive.

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: Development of the One-Way Shear Design Provisions of ACI 318-19 for Reinforced Concrete **Author:** Kuchma, D. A., Wei, S., Sanders, D. H., Belarbi, A., & Novak, L. C. (2019) **Institution**: American Concrete Institute, Structural Journal	• Comprehensive revision of one-way shear design provisions for reinforced concrete in ACI 318-19 building code. • significant advancements in understanding concrete shear resistance, based on multiple databases of beam shear test results spanning multiple decades. • **Motivation**: improve safety, accuracy, and ease of use in shear design.	• Assessing large database of institutions of various test–linear regression.	• Revised ACI 318-19 provisions introduces new general relationship for calculating concrete contribution to shear resistance • Considers factors such as member depth, percentage of longitudinal reinforcement, and axial stress effects. • Shift from previous model-specific approaches to a more unified theory	• Focuses on non-prestressed reinforced concrete members • Future work can include provisions for prestressed concrete members and impacts of factors such as environmental conditions on shear resistance.	• Healthy and substantial, varied dataset to provide a robust model • Scope is for non-prestressed concrete members

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: Database of Shear Experiments on Steel Fiber Reinforced Concrete Beams without Stirrups **Author:** Eva O. L. Lantsoght[1,2] **Institution:** Universidad San Francisco de Quito, Quito, Ecuador[1] Department of Engineering Structures, Civil Engineering and Geosciences, Delft University of Technology, Delft, Netherlands[2]	• Addresses shear capacity of steel fiber reinforced concrete (SFRC) w/o stirrups • International database of 488 experiments • Evaluates various empirical and mechanical prediction models for shear strength. • Underscores challenge of consistent SFRC tensile strength consistently across experimental conditions • Inadequacies of code provisions to fully encapsulate shear behavior of SFRC.	• Assessing large database of international empirical formulas: broad variability.	• None of the empirical formulas accurately describe SFRC behavior w/o stirrups. • Models either underpredict or overpredict shear capacity. None are consistently reliable.	• This is in the context of beams. Although it is highly specific it is a prominent use case. • Need for better SFRC model characteristics in tensile, crack scenarios for shear.	• Comprehensive database. • International implications. • Suggests that empirical models are not the way to go for engineering concrete testing.

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: ANN-Based Shear Capacity of Steel Fiber-Reinforced Concrete Beams without Stirrups **Author:** Miguel Ambabres[1,2], Eva O. L. Lantsoght[3,4] **Institution**: Abambres Lab, Lisbon, Portugal[1] Institute of Education Sciences[2] Universidad San Francisco de Quito, Quito, Ecuador[3] Department of Engineering Structures, Civil Engineering and Geosciences, Delft University of Technology, Delft, Netherlands[4]	• Predicting capacity of steel fiber reinforced concrete (SFRC) w/o stirrups using Artificial Neural Network (ANN)	• 430 data points from various studies. • Multiple ANN models assessed. • Considers: ➢ Fiber volume fraction ➢ Fiber dimensions ➢ Concrete mechanical properties	• ANN achieves perfect results in predicting shear capacity- superior to experiment/ code based models. • MAE of 0.0% across 430 data points.	• Data points were done on small scale. Future work is to incorporate large scale experiments for data points.	• ANN is the way to go over traditional methods. Traditional methods are quasi-mechanical

Paper	Overview	Methodology	Results & Discussions	Research Gap	Strength/ Weaknesses
Title: Novel Adaptive Sampling Algorithm For POD-Based Non-Intrusive Reduced Order Model **Author:** Wang, J., Du, X., & Martins, J. R. R. A. (2021) **Institution**: University of Michigan, Ann Arbor, USA	• A novel adaptive sampling strategy for constructing reduced-order models (ROMs) using Proper Orthogonal Decomposition (POD). • Motivation: improve efficiency and accuracy of non-intrusive ROMs by optimizing the selection of training samples from computational fluid dynamics (CFD) simulations • Airfoil flow field prediction within the transonic regime.	• New adaptive sampling strategy: "conjunction sampling strategy"- a model of models balancing computational budget. • Combines statistical and multicriteria methods to: ➢ Improve predictability ➢ Minimize error by choosing informative training samples (discrete) ➢ 40 training samples	• Conjunction strategy outperformed Latin Hypercube Modeling (LHS) and others. • Significantly reduced standard deviation errors while slightly increasing in mean error (2.6%)	• Study done only on air foils-controlled setting. • 40 data points is small. • This is proof of concept can be tested on other applications. • Scalability of computational effect is not yet known.	• A new strategy for machine learning is available. • It is similar to ensemble learning that incorporates computational criterion.